

Jangjae Lee

Postdoctoral Research Fellow

Department of Civil and Environmental Engineering

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EDUCATION

Ph.D., Civil Engineering

2020 – 2025

Texas A&M University, College Station, TX

Dissertation: Machine Learning (ML)-Based Decision Support: Modeling and Analysis of Power Outages in Texas Under Extreme Climate Conditions

Advisor: Dr. Stephanie G. Paal | Excellence Fellowship (2020–2024) | Dissertation Fellowship (2024–2025)

M.S., Architectural Engineering

2016 – 2018

Hanyang University, Seoul, South Korea

Thesis: Seismic Performance of Exterior Beam-Column Joint Subassemblies Reinforced with Steel Fibers

Advisor: Dr. Chang-Sik Choi | HY-in Scholarship

B.S., Architectural Engineering

2010 – 2016

Hanyang University (ERICA), Ansan, South Korea

Honor Scholarship (2015–2016)

ACADEMIC APPOINTMENTS

Postdoctoral Research Fellow

2025 – Present

University of Houston, Department of Civil and Environmental Engineering, Houston, TX

Principal Investigator (PI): Dr. Abigail L. Beck

Research Intern

May – Aug. 2024

Oak Ridge National Laboratory, Geospatial Science and Human Security Division, Oak Ridge, TN

Mentor: Dr. Sangkeun M. Lee, Senior Staff Data Scientist

RESEARCH INTERESTS

Machine Learning (ML) for Critical Infrastructure Resilience | Power Grid Analytics and Forecasting under Extreme Weather | Climate-Informed Multi-Hazard Risk Modeling | Explainable Artificial Intelligence (AI) for Decision Support | ML for High-Performance Materials | Machine Vision for Infrastructure Inspection | Large Language Models (LLMs) for Emergency Planning | Quantum Computing for Resilience Optimization | Light Detection and Ranging (LiDAR) and Geospatial Data Fusion

PUBLICATIONS

Peer-Reviewed Journal Articles

- [J1] Lee, J., Zhang, Z., & Paal, S. G. (2025). *A data-driven approach to predicting power outages during winter storms in the southern U.S. leveraging nonparametric machine learning models*. *Computational Urban Science*, Springer. <https://doi.org/10.1007/s43762-025-00222-9>
- [J2] Lee, J., Lee, S. M., Chinthavali, S., & Paal, S. G. (2025). *A near-real-time model for predicting electricity disruptions in Texas during winter storms*. *IEEE Access*, 13, 139583–139603. <https://doi.org/10.1109/ACCESS.2025.3596531>
- [J3] Son, D. H., Bae, B. I., Lee, J., & Choi, C. S. (2024). *Shear strength of steel fiber-reinforced concrete (SFRC) exterior beam-column joints with various anchorage details under cyclic loading*. *Structures*, 61, 105940. <https://doi.org/10.1016/j.istruc.2024.105940>
- [J4] Lee, J., Bae, B. I., & Choi, C. S. (2018). *An experimental study on the seismic performance of reinforced concrete (RC) exterior beam-column joints with steel fiber volume fractions*. *Journal of the Architectural Institute of Korea*, 34(4), 15–23. https://doi.org/10.5659/JAIK_SC.2018.34.4.15

Manuscripts Under Review

- [R1] Lee, J., Lee, S. M., Chinthavali, S., & Paal, S. G. (2025). *An integrated model for predicting power outages caused by various types of extreme weather events*. Under review. [R2 Revision]

Manuscripts in Preparation

- [P1] Lee, J., & Paal, S. G. (2026). A knowledge-based predictive model for estimating power outages due to various hazardous weather events. *Manuscript in preparation*.
- [P2] Lee, J., & Beck, A. L. (2026). Community resilience to natural hazards using 311 reports and big data analytics. *Manuscript in preparation*.
- [P3] Lee, J., & Beck, A. L. (2026). U.S. infrastructure quality estimation in coastal regions using Light Detection and Ranging (LiDAR) data fusion. *Manuscript in preparation*.
- [P4] Lee, J., & Son, D. H. (2026). Knowledge transfer predictive model for the shear capacity of steel fiber-reinforced concrete (SFRC) beams. *Manuscript in preparation*.

PRESENTATIONS

Peer-Reviewed Conference Proceedings

- [CP1] Liu, J., Zhang, Z., Cheng, Y., Lee, J., Paal, S., & Li, D. (2025). *A Wide-and-Deep-Based Time Sequence Model for Predicting Power Outages Caused by Extreme Winter Storms*. *International Cartographic Association Proceedings (ICA-Proc)*, 7, 6. <https://doi.org/10.5194/ica-proc-7-6-2025>
- [CP2] Lee, J., Lee, S. M., Paal, S. G., & Chinthavali, S. (2024). *A generalized outage prediction model for various types of extreme climate events in Texas*. *Proceedings of the 2024 Institute of Electrical and Electronics Engineers (IEEE) International Conference on Big Data*, 4162–4166. <https://doi.org/10.1109/BigData62323.2024.10825479>
- [CP3] Lee, J., & Paal, S. G. (2024). *Knowledge transfer predictive models for power outage caused by various types of extreme weather events*. *Proceedings of the 2024 IEEE International Conference on Big Data*, 8227–8229. <https://doi.org/10.1109/BigData62323.2024.10825465>

Refereed Oral Presentations

- [O1] Lee, J., & Beck, A. L. (2026, June). *Automated structural assessment and tilt measurement of coastal utility poles: Leveraging human-refined Grounded Segment Anything Model (SAM) 3 and Light Detection and Ranging (LiDAR) integration*. *American Society of Civil Engineers (ASCE) Engineering Mechanics Institute (EMI) Conference*, Boulder, CO. [Accepted]
- [O2] Lee, J., & Paal, S. G. (2025, May). *Harnessing transfer learning to predict power outages from diverse climate hazards*. *ASCE EMI Conference*, Anaheim, CA.
- [O3] Lee, J., & Paal, S. G. (2024, May). *A generalized model for predicting power outages in Texas during extreme weather events: Integrating lagged information, geographical, climatic, and socio-demographic data*. *EMI/Probabilistic Mechanics Conference (PMC)*, Chicago, IL.
- [O4] Lee, J., & Paal, S. G. (2024, May). *A generalized machine learning model for predicting power outages across multiple climate zones*. *ASCE EMI Conference*, Chicago, IL.
- [O5] Lee, J., & Paal, S. G. (2023, May). *Ensemble-based time series analysis considering lag information and feature importance to predict power outages during winter storms*. *Engineering Mechanics Institute (EMI) Conference*, Atlanta, GA.

Posters and Lightning Talks

- [PT1] Lee, J., & Beck, A. L. (2026, January). *Integrated spatiotemporal assessment of infrastructure resilience and community vulnerability in Houston*. *Natural Hazards Engineering Research Infrastructure (NHERI) Computational Symposium*, Berkeley, CA. [Accepted]
- [PT2] Lee, J., Lee, S. M., & Paal, S. G. (2025, January). *An integrated model for predicting power outages during various types of extreme weather events in Texas*. *NHERI Computational Symposium*, Los Angeles, CA. [Poster]
- [PT3] Lee, J., & Paal, S. G. (2024, January). *Ensemble-based time series modeling for predicting power outages during extreme weather: A multi-factor approach integrating meteorological, geographical, and socio-demographic features*. *NHERI Computational Symposium*, Los Angeles, CA. [Lightning Talk]

Additional Conference Publications (Korea, 2016–2018)

Seven peer-reviewed conference papers on seismic performance and fiber-reinforced concrete (Korea Concrete Institute (KCI), CODE, Architectural Institute of Korea (AIK)). Full list available upon request.

RESEARCH EXPERIENCE

Postdoctoral Research Fellow

2025 – Present

University of Houston, Houston, TX | PI: Dr. Abigail L. Beck

- **Community Resilience:** Developing data-driven frameworks integrating 311 service requests and socio-demographic data to quantify urban resilience and infrastructure outage impact across Houston neighborhoods.
- **Machine Vision:** Implementing Grounded Segment Anything Model (SAM) for automated assessment of utility pole deterioration and tilt angle measurement via Light Detection and Ranging (LiDAR) point cloud integration.
- **Infrastructure Assessment:** Investigating U.S. coastal infrastructure quality through statistical modeling and empirical fieldwork validation.

Doctoral Researcher

Jan. 2022 – Aug. 2025

Texas A&M University, College Station, TX | Advisor: Dr. Stephanie G. Paal

- Architected Machine Learning (ML) frameworks for large-scale power outage prediction across nine U.S. states using multi-source meteorological and socio-demographic datasets.
- Developed and optimized ensemble models (XGBoost, LightGBM, CatBoost) with Bayesian hyperparameter tuning and SHapley Additive exPlanations (SHAP)-based explainability.
- Deployed near-real-time operational prediction systems; applied transfer learning for data-sparse climate hazard scenarios.

Research Intern

May – Aug. 2024

Oak Ridge National Laboratory, Oak Ridge, TN | Grid Resilience Research Group | Mentor: Dr. Sangkeun M. Lee

- Conducted national-scale power grid resilience analysis; developed interpretable ML models for critical energy infrastructure; contributed to one journal article and one conference paper.

Hands-On Research

Aug. – Dec. 2020

Texas A&M University, College Station, TX | Advisor: Dr. Stephanie G. Paal

- Implemented Convolutional Neural Network (CNN)-based steel defect detection via ResNet50 transfer learning; evaluated with Intersection over Union (IoU), Precision, Recall, and F1-Score.

Graduate Research Assistant and Mentor

Nov. 2016 – Feb. 2021

Hanyang University, Seoul, South Korea | Advisor: Dr. Chang-Sik Choi | National Research Foundation (NRF)-funded (\$250,000)

- Conducted experimental research on seismic performance of Steel Fiber-Reinforced Concrete (SFRC) structures; designed and fabricated specimens and performed cyclic loading tests.
- Supported National Research Foundation (NRF) project proposal development; mentored junior graduate students on experimental protocols and data acquisition.
- Evaluated Ultra-High-Performance Cementitious Composite (UHPCC) shear bond strength; received **Grand Prize** at the Korea Institute of Building Construction (KIBC) Design Contest.

Undergraduate Research Assistant

Apr. – Oct. 2015

Hanyang University (ERICA), Ansan, South Korea | Advisor: Dr. Han-seung Lee

- Conducted capstone research on shear bond strength evaluation of Ultra-High-Performance Cementitious Composite (UHPCC) using varying surface treatment methods.

TEACHING EXPERIENCE

Guest Lecturer

Spring 2026

University of Houston | Instructor of Record: Dr. Vedhus Hoskere

ENGR 2301: Engineering Mechanics I (Statics)

- **3D Rigid Body Equilibrium:** Support reactions for statically determinate 3D systems; modeled constraints and formulated equilibrium equations using Cartesian vector notation.
- **Truss Analysis:** Method of Joints for 3D space trusses; determinacy criteria and identification of zero-force members in complex spatial configurations.

CIVE 3337: Structural Analysis | Instructor of Record: Dr. Abigail L. Beck

- **Structural Classification and Stability:** Taught determinacy using the r vs. $3n$ relationship; trained students to identify unstable geometric and static configurations in truss and frame systems.

Guest Lecturer

Fall 2025

University of Houston | ENGR 2301: Engineering Mechanics I (Statics) | Instructor of Record: Dr. Vedhus Hoskere

- **Centroids and Moment of Inertia:** Designed and delivered lecture for 40+ undergraduate students; integrated physical demonstrations (ruler-bending for Second Moment of Area; pen-balancing for Center of Gravity) to bridge mathematical derivations with engineering intuition.

Teaching Assistant

Spring 2023 – Spring 2024

Texas A&M University | Zachry Department of Civil and Environmental Engineering

CVEN 363: Engineering Mechanics — Dynamics | Instructor: Dr. Stephanie G. Paal

- Facilitated biweekly recitation sessions for 60+ students across two sections; clarified kinematics and kinetics concepts.
- Developed solution keys and grading rubrics; held weekly office hours and organized pre-exam review sessions.

CVEN 345: Theory of Structures | Instructors: Dr. J. Mander, Dr. L. Lowery, Dr. S. Hurlbaas

- Managed grading and assessment for 150+ students in collaboration with a team of three instructors; created supplemental study materials on structural analysis methods.

PROFESSIONAL SERVICE

Manuscript Review

- **Ad hoc reviewer**, *Reliability Engineering and System Safety* (Elsevier)
- **Ad hoc reviewer**, *Natural Hazards Review* (ASCE)
- **Ad hoc reviewer**, *International Journal of Disaster Risk Reduction* (Elsevier)

Professional Memberships

- Member, Natural Hazards Engineering Research Infrastructure (NHERI) Graduate Student Council (2024–2025)
- Member, American Society of Civil Engineers (ASCE) (2016–present)
- Member, American Concrete Institute (ACI) (2016–present)
- Member, Architectural Institute of Korea (AIK) (2016–present)
- Member, Korea Concrete Institute (KCI) (2016–present)

STUDENT MENTORSHIP

University of Houston

2025 – Present

- Mentoring four graduate research assistants on research methodology, data analysis, and technical writing:
 - Olubunmi Ogunleye, Nafeezat Ajenifuja, Swastika Barua, Roydon Jude Samudio

Texas A&M University

2022 – 2025

- Mentored one graduate student (Jacob Murphy) and one undergraduate student (Robyn Andrew) through the full research lifecycle, including experimental design, analysis, and scientific writing.

Hanyang University

2018 – 2021

- Supervised two junior researchers (Jin-Sung Kim, Ki-Hyeon Kim) on National Research Foundation (NRF)-funded experimental protocols, seismic performance testing, and data acquisition.

HONORS AND AWARDS

Dissertation Fellowship — Texas A&M University	2024 – 2025
Excellence Fellowship — Texas A&M University	2020 – 2024
HY-in Scholarship — Hanyang University (Seoul)	2016 – 2018
Grand Prize — 10th Construction Technology Competition, Korea Institute of Building Construction	2015
Honor Scholarship — Hanyang University (ERICA)	2015 – 2016
Bronze Prize — Capstone Design Competition, Hanyang University (ERICA)	2015

TECHNICAL SKILLS

3D Point Cloud and Machine Vision: CloudCompare, Grounded Segment Anything Model (SAM), Semantic Segmentation, Point Cloud Processing, Light Detection and Ranging (LiDAR) Data Analysis

Artificial Intelligence and High-Performance Computing: Python (scikit-learn, Pandas, NumPy, GeoPandas), TensorFlow, PyTorch, High-Performance Computing (HPC) — Parallel Computing, MATLAB, R

Machine Learning Techniques: Ensemble Methods (XGBoost, CatBoost), Transfer Learning, Deep Learning (Convolutional Neural Network (CNN), Recurrent Neural Network (RNN)), Bayesian Optimization, SHapley Additive exPlanations (SHAP)

Engineering and Geospatial Software: SAP2000, ABAQUS, AutoCAD, Revit, QGIS, ArcGIS

General: LaTeX, Adobe Photoshop, Microsoft Office Suite

WORK EXPERIENCE

Design Assistant (Part-time) June – Aug. 2014
In-dong Architectural Firm, Seoul, South Korea

- Assisted in building design competition concept development; produced SketchUp 3D models, AutoCAD floor plans and elevations, and physical foam board presentation models.

Teaching Assistant (Part-time) May – Dec. 2015
PK Dok-hak Jae-su Academy, Seoul, South Korea

- Delivered subject-matter lectures, managed classroom administration, and developed and graded examination materials.

REFERENCES

Stephanie G. Paal, Ph.D.

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